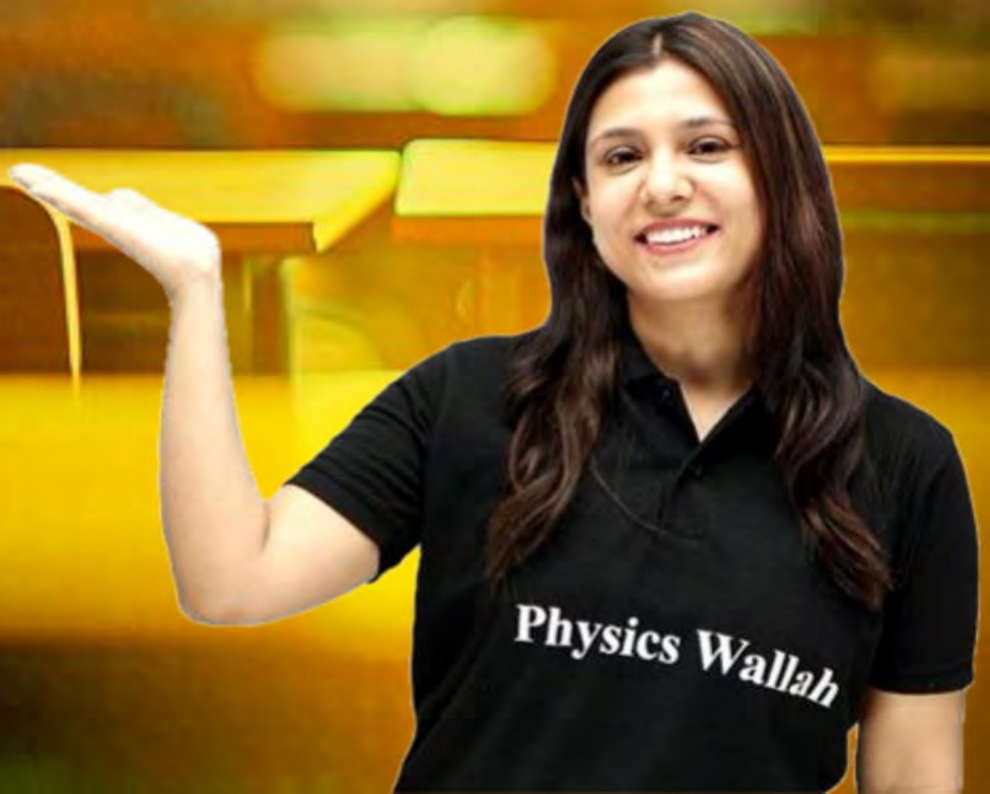
RADIANT 2026

Biology

Digestive System

Lecture – 01

By - Upasana Mongia Ma'am



Topics to be covered

1. Parts of Digestion system
2. Alimentary Canal
3. Mouth and Teeth & its types
4.

Ninja ok !

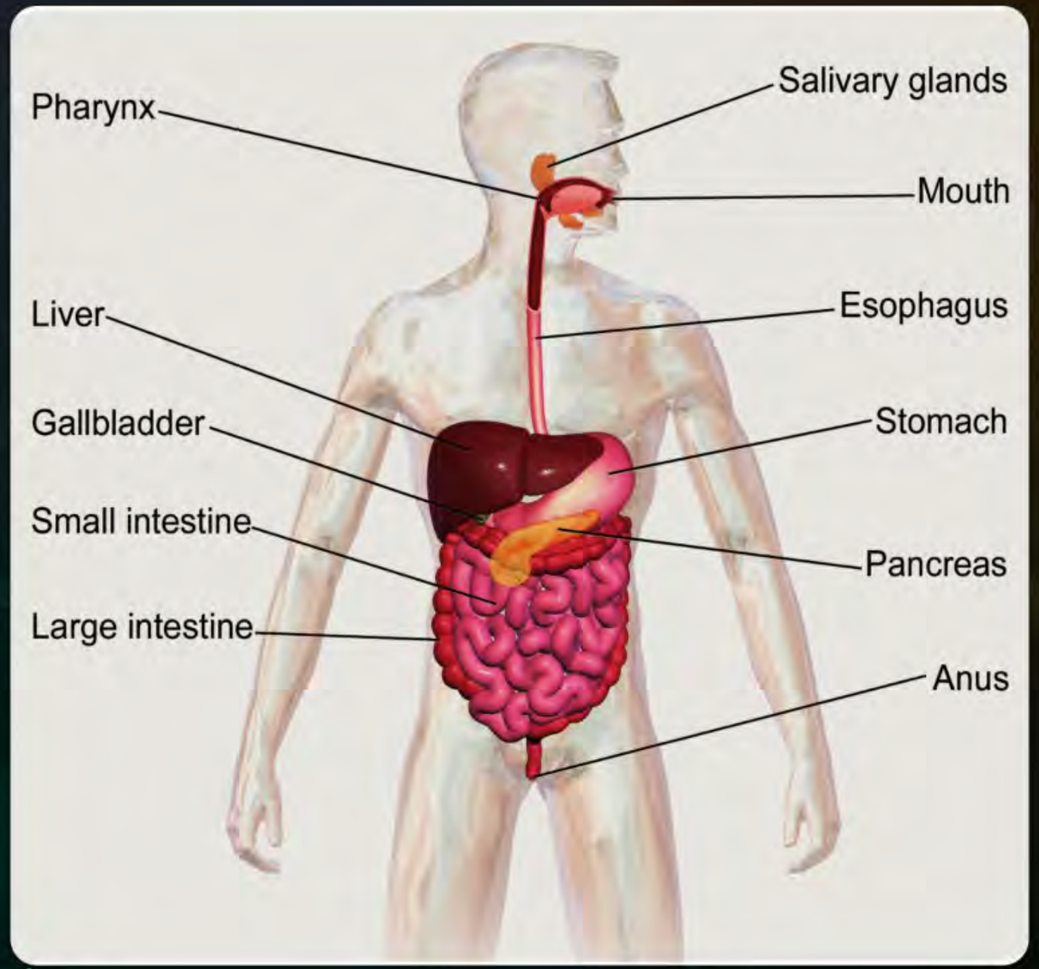


Digestion

- Digestion is the process of breakdown of naturally occurring foodstuffs into a diffusible form.
- It is a process which makes the foods soluble and of such a chemical nature that they can be absorbed through living membranes.

- The enzymes which help in the process of digestion are called digestive enzymes.

* They work on fixed pH & temperature
* They act on specific substrate like a Catalyst.



G-P.A-ATP

cell

Laccha Parantha

Chana

Digestion करदो

large Complex Molecules को Break करदो

small Simple Molecules.

Nutrients

→ Carbohydrates → Glucose

→ Proteins → Amino acids

→ fats → Fatty acids & Glycerol.

Digestion with Enzymes.

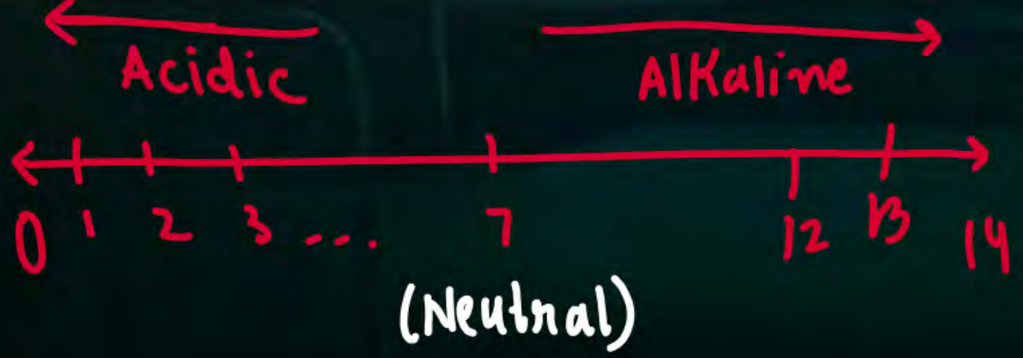
Enzyme

Carbohydrates → Glucose.

Substrate — Product

Proteins → Amino acids.

Fats → Fatty acids + Glycerol.



37°C

35°C - 40°C

optimum temp. for enzymes.

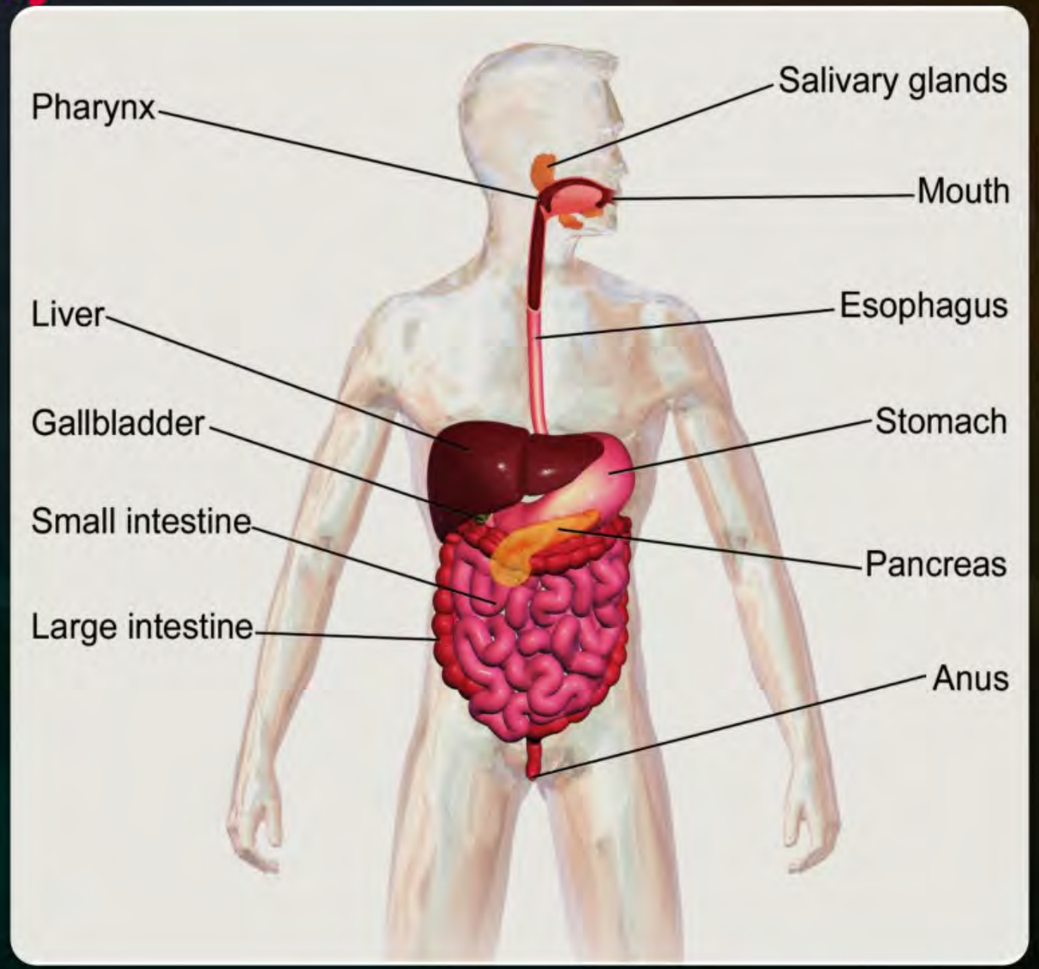
Digestive System

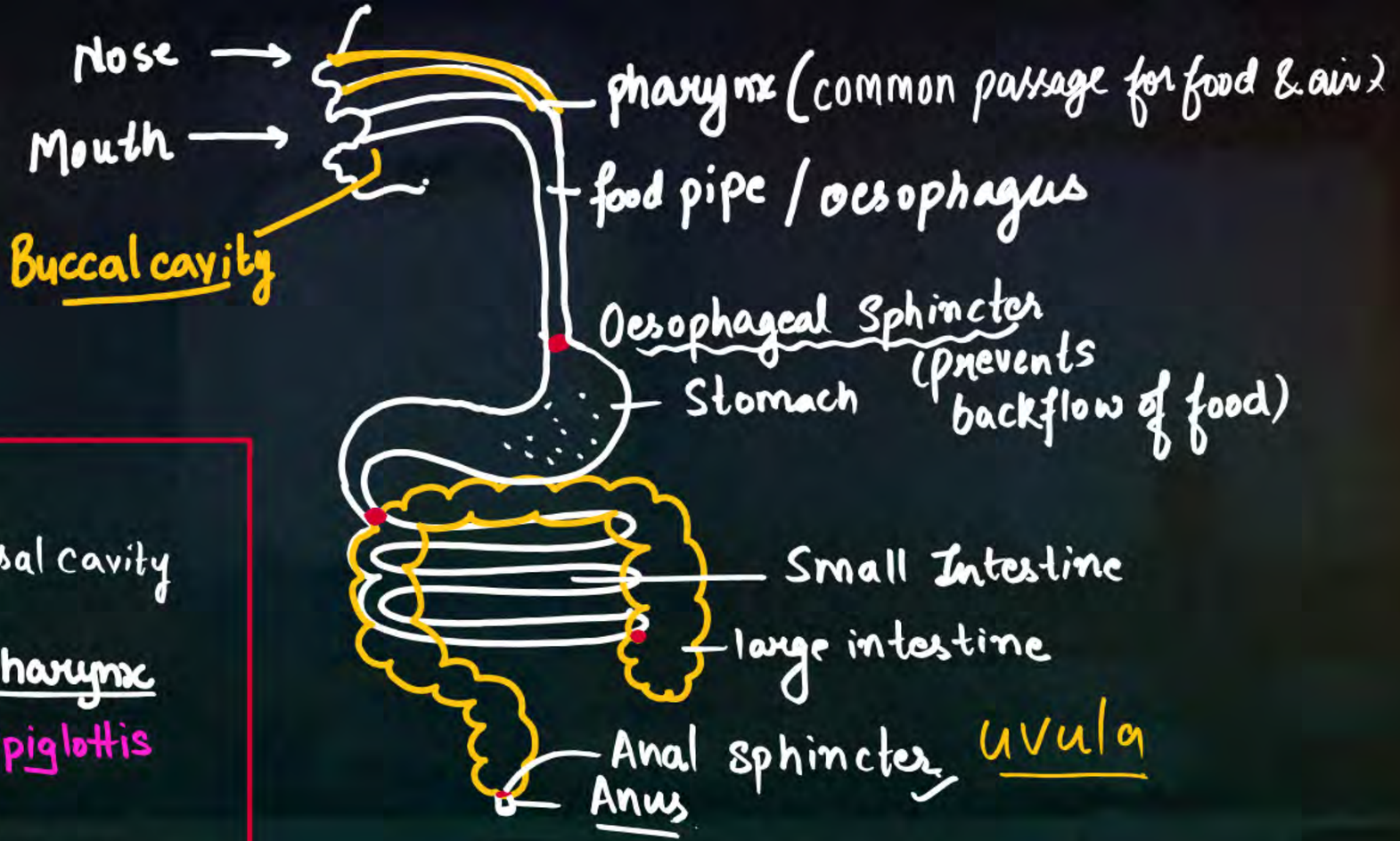
Organ System
↳ (group of organs)

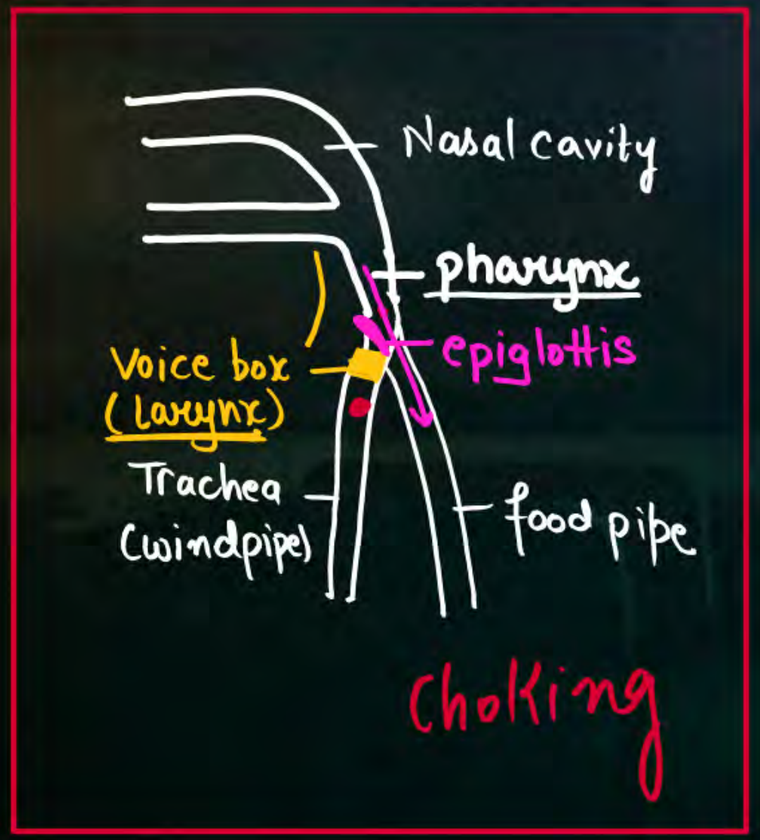
Alimentary canal

- 9m long
- Muscular tube
- Starts with mouth ,ends with anus
- Highly coiled in some regions

Digestive glands







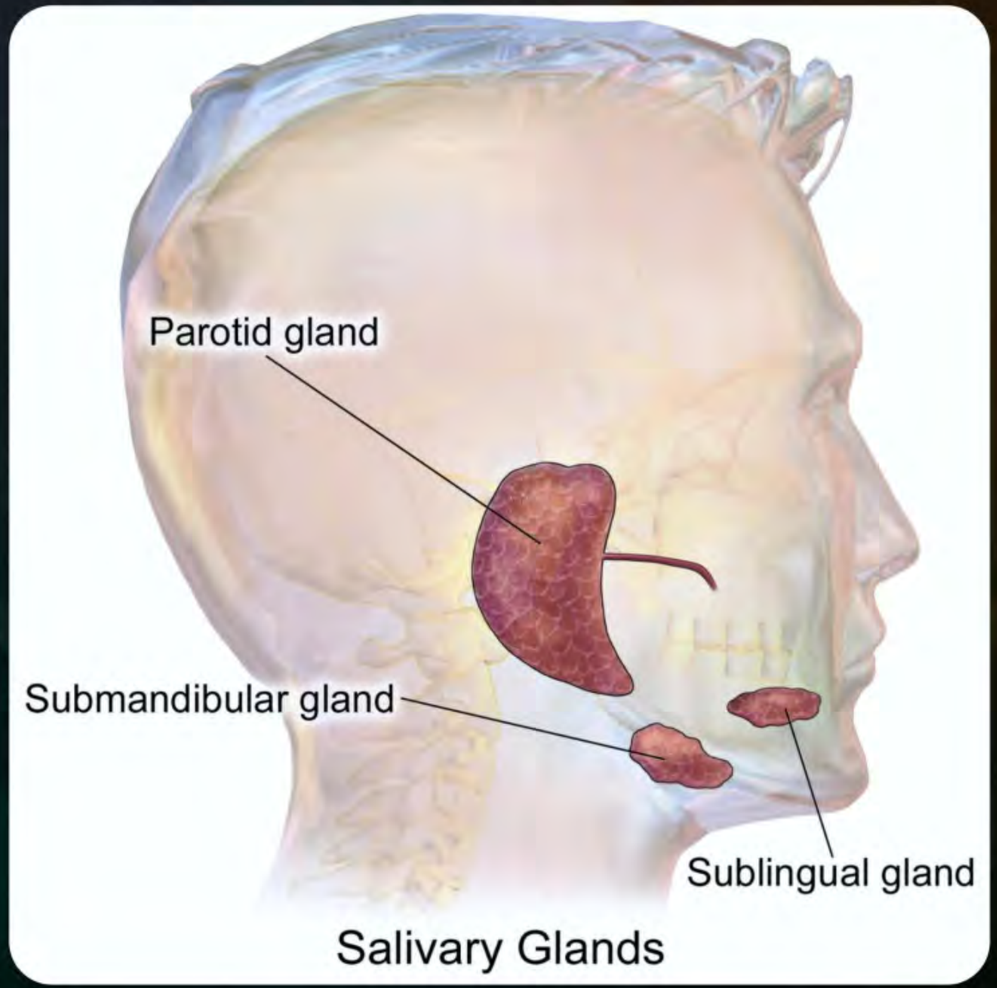


Epiglottis

→ It is a flap like structure which covers glottis (upper part of Trachea) during swallowing.

Mouth

- The space where the food is chewed and mixed with the saliva is called the mouth or mouth cavity./Buccal Cavity.
- The mouth opens into the buccal cavity which has teeth, a tongue and three pairs of salivary glands.



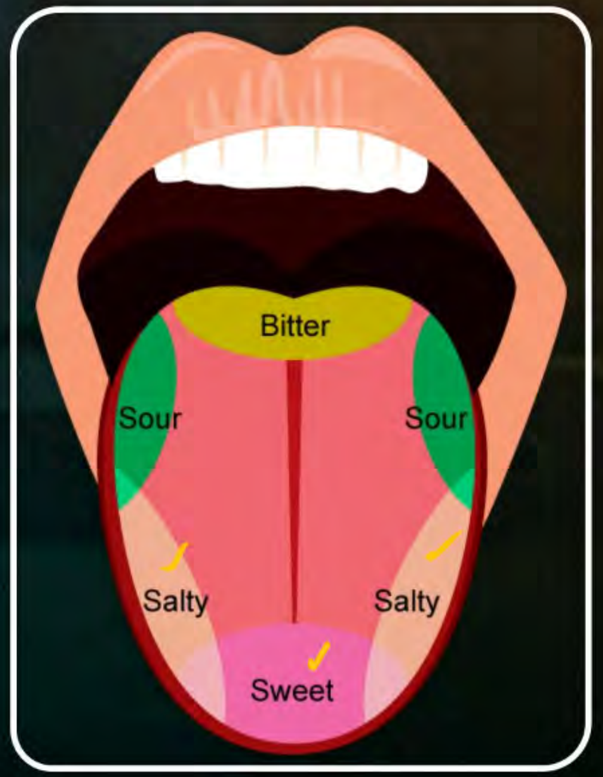
Salivary Glands

Functions of the Mouth

- During ingestion, food is taken in through the mouth.
- Lips help in closing the mouth and sucking and sipping , speaking,sensations

Tongue

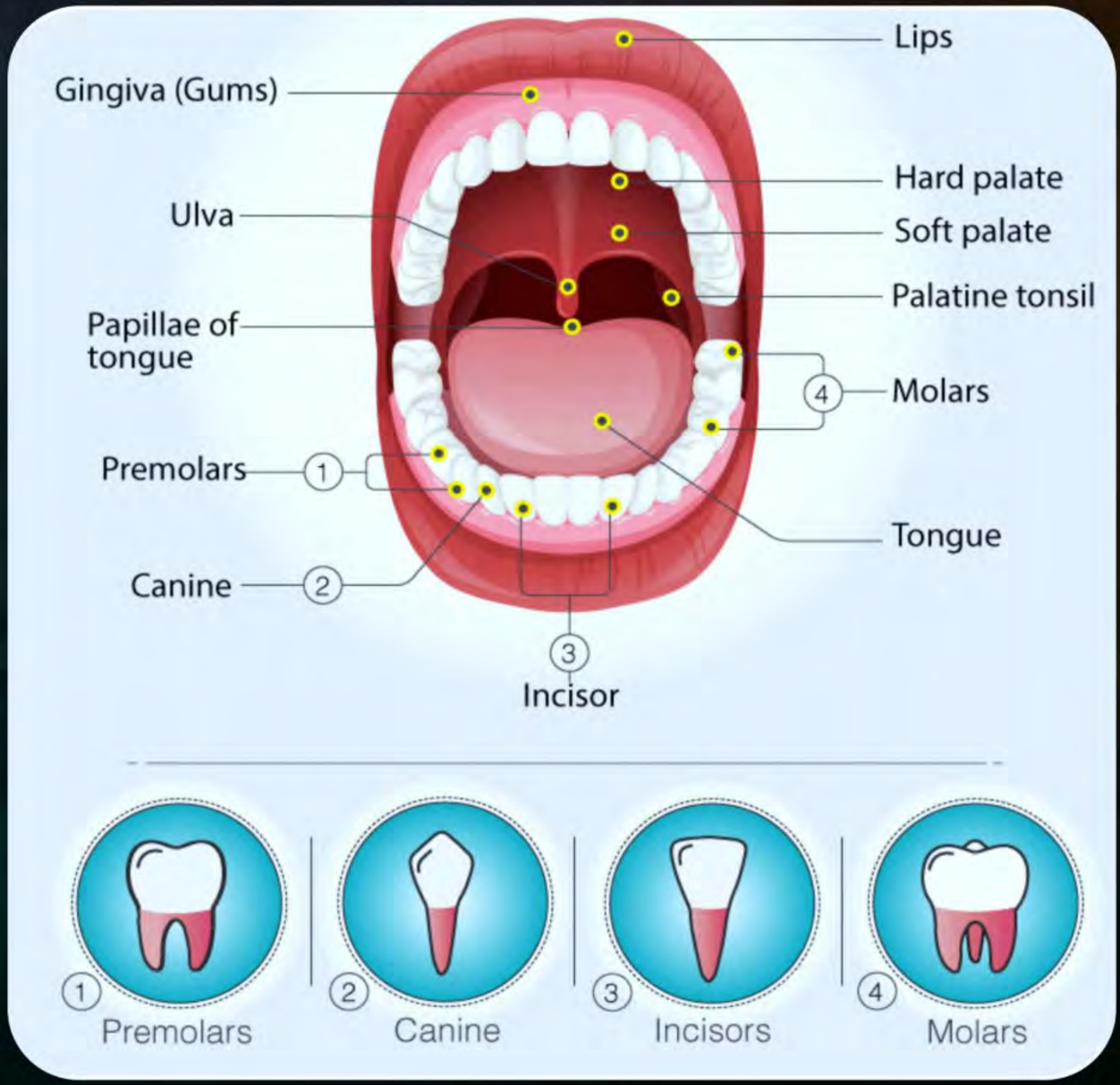
- The tongue is a fleshy muscular organ attached to the floor of the buccal cavity at the back region of the mouth.
- The tongue helps in mixing of saliva with the food while chewing.
- It also helps in mastication and swallowing of food.
- The tongue has several taste buds at its tip which detects different tastes of food.



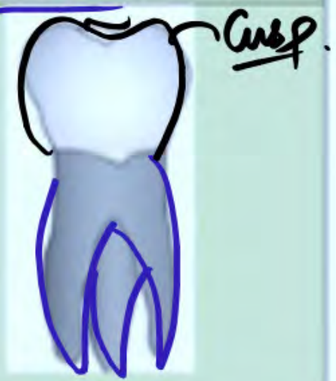
Teeth

- The upper and lower jaws are embedded with teeth.
- Teeth of different shapes are called heterodont.
- Example: Human teeth.
- Teeth which are similar to each other are called homodont.
- Examples Teeth of lizard and frog.





Types of human teeth

	Incisor	Canine	Premolar	Molar
				
Position in mouth	Front	Either side of incisors	Behind canine	Back
Description	Chisel-shaped (sharp edge)	Slightly more pointed than incisors	2 points (cusps), 1 or 2 roots	4 or 5 cusps 2 or 3 roots
Function	Biting of pieces of food	Similar function to incisors (Tearing)	Tearing and grinding food	Chewing and grinding food



Dental formula



$\frac{2102}{2102}$ $\frac{5}{5}$ ×2 $\frac{10}{10}$

Child [20]
(milk teeth / temporary)

$\frac{2122}{2122}$

Teenager. [28 teeth]

Incisors	Canines	PreMolars	Molars.
2	1	2	3
2	1	2	3

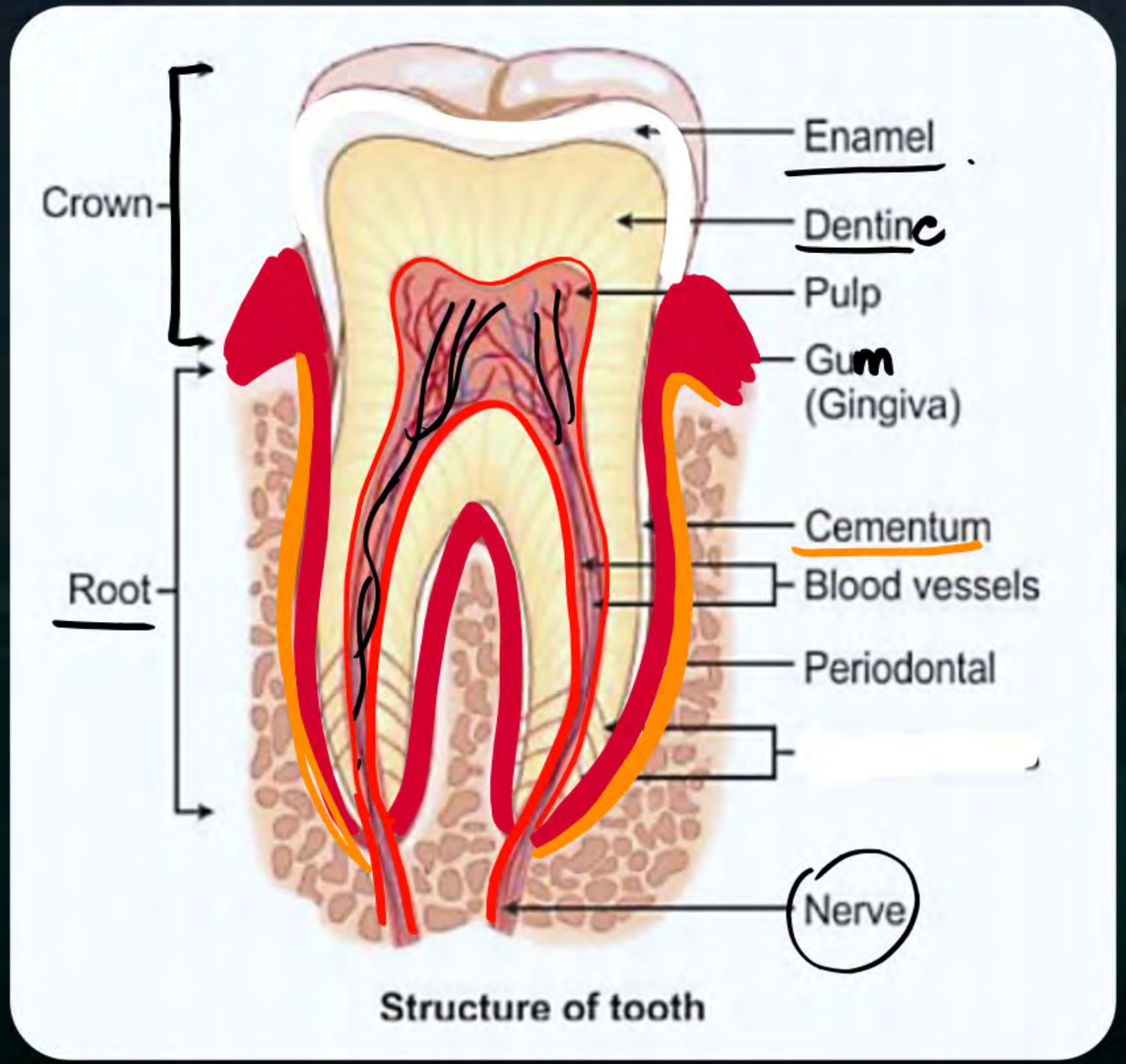
Dental Formula
[Human Adult]
32

Functions of Teeth

- Teeth help to cut food into small pieces, to chew and to grind it.
- They prepare food for digestion by mastication.
- They help in speaking and add to facial beauty.



Structure of a Tooth



Structure of tooth

Structure of a Tooth

- **Enamel** – hardest substance of the body,covers the crown.
- **Dentine** –forms bulk of tooth,harder then bone but less than enamel, has place for pulp cavity.
- **Cement** – bone like structure covers and fix the root.
- **Pulp** – soft connective tissue, consists of blood capillary, lymph vessels, nerve fibres.

Question

The human dental formula is

A $\frac{2,1,2,0}{2,1,2,0}$

B $\frac{2,1,2,2}{2,1,2,2}$

C $\frac{2,1,1,0}{2,1,1,0}$

D $\frac{2,1,2,1}{2,1,2,1}$

Question

Which is the correct sequence of body parts in the human alimentary canal?

A. Mouth → stomach → small intestine → large intestine → oesophagus

B. Mouth → oesophagus → stomach → small intestine → large intestine

C. Mouth → stomach → oesophagus → small intestine → large intestine

D. Mouth → oesophagus → stomach → large intestine → small intestine

Q. After digestion the Carbohydrates are broken into-

A) Peptides

B) Maltose & Glucose ✓

C) Glycerol

D) All of the above

Disaccharide

Maltose → Glucose + Glucose


Thank You